

Shreshta Agrawal

2nd Year Undergraduate | Indian Institute of Technology Kharagpur
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EDUCATION

B.Tech in Aerospace Engineering, IIT Kharagpur	Aug 2024 – May 2028	CGPA- 9.00
Class X (CBSE), Orchids International School, Thane	Apr 2021 – Mar 2022	Grade – 98%
Class XII (HSC), Pace Junior Science College, Dadar	Apr 2023 – Mar 2024	Grade – 93.67%

RESEARCH INTERESTS

Aerodynamics, Computational Fluid Dynamics (CFD), Vortex and Unsteady Flow Dynamics, Fluid-Structure Interaction, Computational Aeromechanics, Finite Element Analysis (FEA), Physics informed Neural Networks (PINN), Deep Learning for Flow Prediction

EXPERIENCE

Research and Development Intern Indian Institute of Technology Kharagpur Prof. S. M. Dash	<i>Jan 2026 - Present</i>
- Operated MiniLDV with MSE data acquisition software for high-precision velocity measurements and flow characterization. - Utilised a micro-traverse positioning mechanism for precise probe alignment and controlled spatial sampling within the test section. - Designed a circular cylinder wake flow experimental rig, including CAD modelling and fabrication coordination for custom test hardware. - Developing an experimental platform to investigate vortex shedding and bluff-body wake flow structures in controlled laboratory conditions.	
Research and Development Intern Indian Institute of Technology Kharagpur Prof. Prasun Jana	<i>Mar 2026 - Present</i>
- Developed PINN models for projectile motion with drag, enabling forward trajectory prediction and inverse parameter estimation. - Estimated physical parameters including initial velocity, launch angle, and drag coefficients from observed motion data. - Evaluated model robustness under non-ideal effects such as stochastic disturbances, drag variability, and external forcing. - Extended the framework to UAV flight dynamics with cross-winds, vibration effects, and aerodynamic forces as physics constraints.	

PROJECTS

Fixed-Wing RC Aircraft with Payload Deployment System Group Project IIT Kharagpur	<i>Dec 2025 - Feb 2026</i>
<i>Objective: Design and development of a remotely controlled aircraft with mid-air payload deployment capability.</i> - Designed and fabricated a remote-controlled fixed-wing aircraft with an integrated payload release mechanism for controlled mid-air deployment. - Developed CAD models of airframe components and payload housing while optimising centre-of-gravity placement and wing loading for stable flight. - Engineered a servo-actuated payload release system integrated with onboard radio control architecture for remotely triggered deployment. - Performed airframe assembly, subsystem integration, and flight testing to evaluate aerodynamic stability, control response, and flight performance.	
Unsteady Aerodynamics of Flapping Wing Mechanisms Academic Project IIT Kharagpur	
<i>Objective: Investigation of vortex dynamics and unsteady lift generation in flapping wings.</i> - Conducted a literature-based investigation of vortex shedding mechanisms and unsteady wake dynamics in flapping wing aerodynamics. - Analysed formation, evolution, and stability of leading-edge vortices (LEVs) and their role in delayed stall and aerodynamic lift enhancement. - Examined counter-rotating vortex pair (CRVP) formation and complex vortex–wake interactions in oscillating flapping wing kinematics.	

RELEVANT COURSEWORK

- Mathematics:** Linear Algebra, Transform Calculus, Applied Computational Methods, Essentials of Machine Learning
- Aerodynamics:** Introduction to Aerodynamics, Low Speed Aerodynamics, Aerodynamics Laboratory
- Structures:** Introduction to Aerospace Structures, Structures Laboratory
- Controls & Dynamics:** Introduction to Flight Vehicle Controls, Engineering Dynamics

HONORS AND AWARDS

First Prize, Boeing Aeromodelling Competition (Zonal Level)
Awarded ₹50,000 cash prize for designing and flying a payload-dropping fixed-wing UAV, outperforming competing teams across mission accuracy and flight performance.

TECHNICAL SKILLS

- Languages & Frameworks:** Python (Pandas, NumPy, TensorFlow, PyTorch, Matplotlib) | C/C++
- Softwares:** Matlab | Simulink | Ansys Fluent | LS Dyna | Solidworks | Fusion 360 | Onshape | Jupyter Notebooks | VS Code | Adobe Illustrator | MSE 1D acquisition manager | MicroMotion Traverse Control Software
- Certifications:** Machine Learning Specialisation- Andrew Ng, Stanford University

POSITION OF RESPONSIBILITY

Awaaz, IIT Kharagpur Senior Reporter Kharagpur, West Bengal	<i>July 2025 – May 2026</i>
- Led on-ground reporting on campus issues, conducting interviews and coordinating with editorial teams for timely publications. - Contributed to reporting that gained external recognition, with Awaaz featured in The Times of India, The Telegraph, and The Sunday Times.	